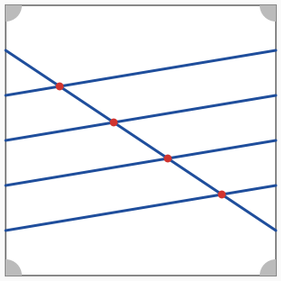
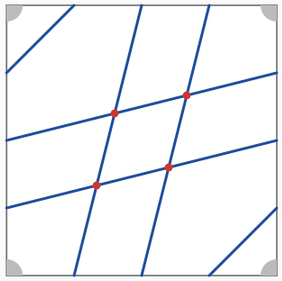
4 self-intersections: 29 orbits of unoriented curves, 27 primitive

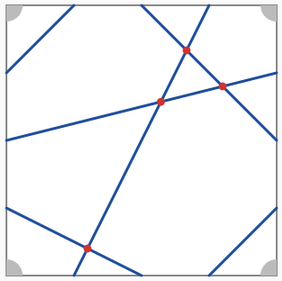
Opposite sides are glued; the puncture is at the corners. a exits right and re-enters left, b exits top and re-enters bottom.



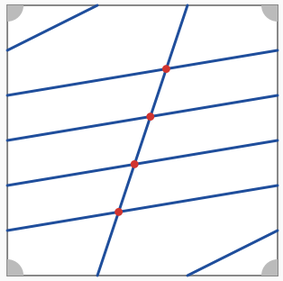
1. aaaaa



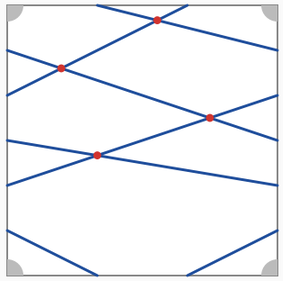
2. aaabbb



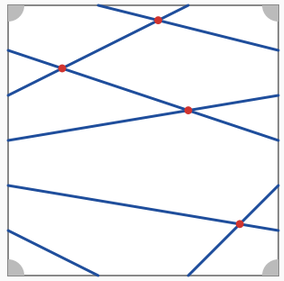
3. aabbaB



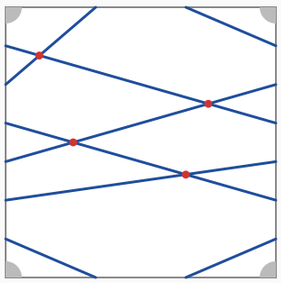
4. aaaaabb



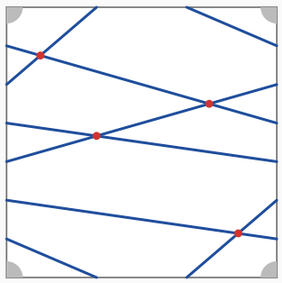
5. aaaabaB



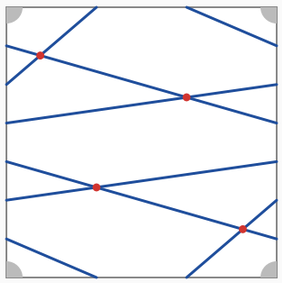
6. aaabaaB



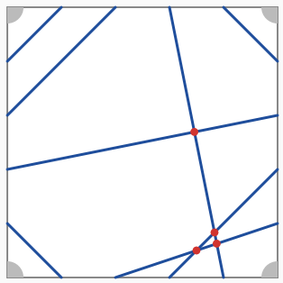
7. aaaaabAB



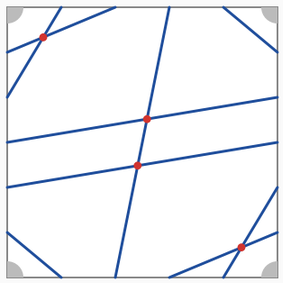
8. aaaabAAB



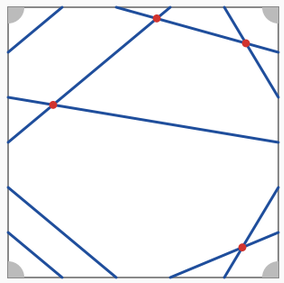
9. aaabAAAB



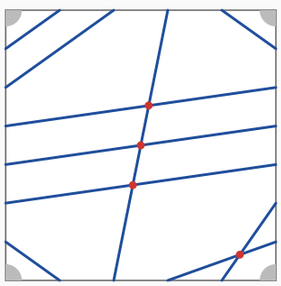
10. aabaBAbb



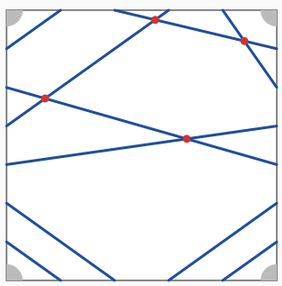
11. aaabABBAb



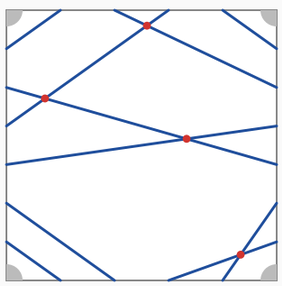
12. aabaBAbAB



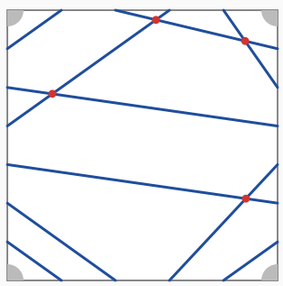
13. aaaabbaBAb



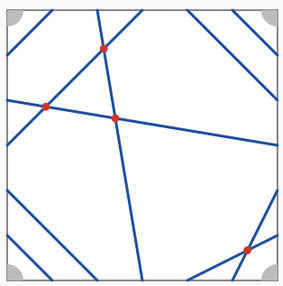
14. aaabaBabAB



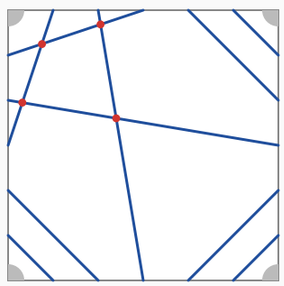
15. aaabaBAbaB



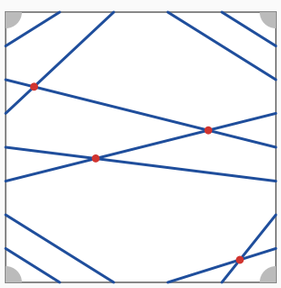
16. aabaaBabAB



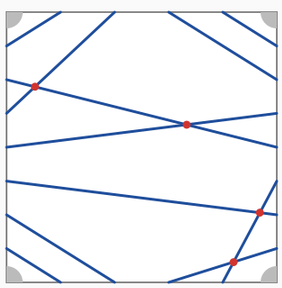
17. aabbABabAB



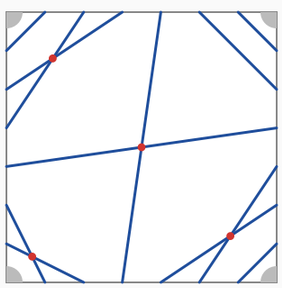
18. aabABabbAB



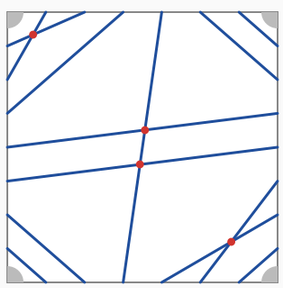
19. aaaabABabAB



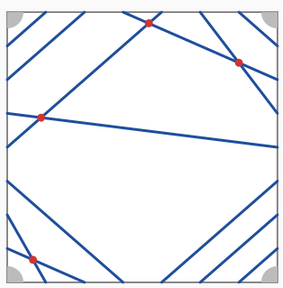
20. aaabAABabAB



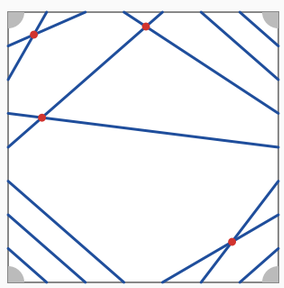
21. aabABabABBAb



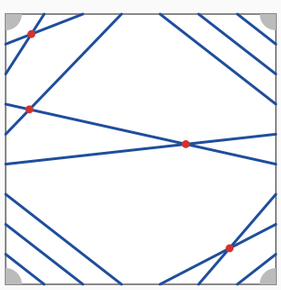
22. aaabbaBAbaBAb



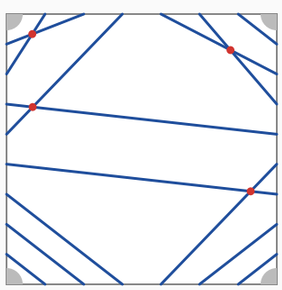
23. aabaBabABabAB



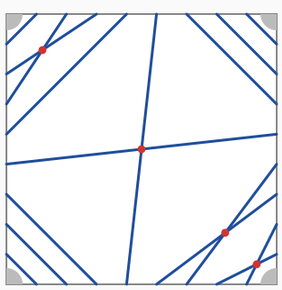
24. aabaBAbaBAbaB



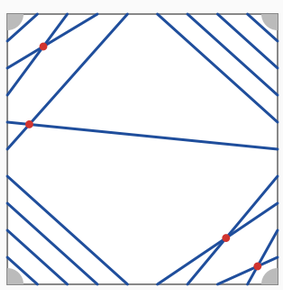
25. aaabABabABabAB



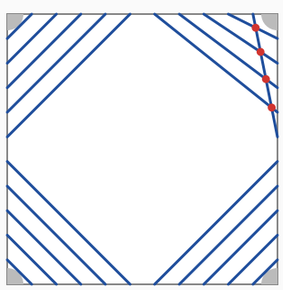
26. aabAABabABabAB



27. aabbaBAbaBAbaBAb



28. aabABabABabABabAB



29. abABabABabABabABabAB